# Handling Arguments in Functions

Positional, keyword, default, *args, **kwargs, plus the `/` and `*` markers — and how Python really passes objects.



## Parameter kinds & their order

```
def f(a, b, /, c, d=4, *args, e, g=6, **kwargs):
    return a, b, c, d, args, e, g, kwargs

print(f(1, 2, 3, 40, 50, 60, e=5, x=7))
```

OUTPUT
```
(1, 2, 3, 40, (50, 60), 5, 6, {'x': 7})
```

| Part | Meaning |
|---|---|
| `a, b` (before `/`) | positional-only |
| `c, d=4` | positional-or-keyword (`d` has a default) |
| `*args` | extra positional → tuple |
| `e, g=6` (after `*args`) | keyword-only (`e` is required) |
| `**kwargs` | extra keyword → dict |

## Bare `/` and `*` markers

```
def pay(amount, /, *, currency="INR"):
    return f"{amount} {currency}"

print(pay(250, currency="USD"))
try:
    # positional-only passed by name
    pay(amount=250)
except TypeError:
    print("TypeError")
```

OUTPUT
```
250 USD
TypeError
```

## Unpacking in calls

```
def vol(l, w, h): return l * w * h
dims = (2, 3, 4)
spec = {"l": 2, "w": 3, "h": 4}
print(vol(*dims), vol(**spec))
```

OUTPUT
```
24 24
```

## The mutable-default trap

```
# ONE list shared by all calls
def add_bad(x, bucket=[]):
    bucket.append(x)
    return bucket

def add_ok(x, bucket=None):          # safe pattern
    if bucket is None:
        bucket = []
    bucket.append(x)
    return bucket

print(add_bad(1), add_bad(2))
print(add_ok(1), add_ok(2))
```

OUTPUT
```
[1, 2] [1, 2]
[1] [2]
```

## Pass by object reference

```
def mutate(lst):
    lst.append(99)        # changes the caller's list
def rebind(lst):
    lst = [0]             # rebinding stays local

data = [1]
mutate(data); rebind(data)
print(data)
```

OUTPUT
```
[1, 99]
```

## Forwarding arguments

```
def logged(fn, *args, **kwargs):
    print("calling", fn.__name__, args, kwargs)
    return fn(*args, **kwargs)

print(logged(pow, 2, 5))
```

OUTPUT
```
calling pow (2, 5) {}
32
```

## Common errors

| Message (TypeError) | Cause |
|---|---|
| missing 1 required positional argument | too few arguments |
| takes 2 positional arguments but 3 were given | too many arguments |
| got multiple values for argument 'x' | positional + keyword clash |
| got an unexpected keyword argument | typo, or no `**kwargs` |

**TIP** Use keyword-only parameters for flags: `save(data, *, overwrite=False)` forces `save(d, overwrite=True)` – no mystery booleans.